Data Processing Architectures: Batch and Streaming in Database Systems

This dissertation aims to provide a comparative analysis of batch processing and streaming processing within modern data management systems, with a particular focus on database components and analytics-oriented architectures. The study is conducted in the context of the medical domain, using simulated hospital-type data designed to reflect realistic operational scenarios of healthcare information systems.

The project involves the design and implementation of a data warehouse intended for the storage and analysis of structured data such as patient admissions, consultations, medical procedures, and operational indicators. The database structure is designed to support both historical data analysis and near–real-time data processing, offering a unified framework for comparing the two processing paradigms.

Batch processing is examined through periodic data loading and aggregation workflows, optimized for large volumes of historical data and complex analytical queries. In parallel, streaming processing is evaluated through continuous data ingestion and real-time updating of analytical results, highlighting the advantages and limitations of each approach with respect to latency, scalability, and system complexity.

By implementing both approaches on the same dataset, the dissertation provides a practical perspective on the criteria for choosing between batch and streaming architectures. It emphasizes the impact of database design and processing strategies on the analytical capabilities of information systems, particularly in the medical domain.

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